Activity: Compare Symmetric vs. Asymmetric Encryption

Feature	**Symmetric Encryption**	**Asymmetric Encryption**
Number of Keys	One single private	Two keys, one private, one public
Speed	Faster	Slower
Use Case Example	AES–Advanced Encryption Standard	RSA–Rivest Shamir Adleman
Key Management	Not really used in this instance	Better here because it is used to exchange keys, and digital signatures
Best For	Speed, and large data sets, databases	Key, an digital signature exchange
Main Challenge	Securely sharing the key between the sender and receiver	It is slower, and requires more computational resources

Symmetric encryption is ideal for large data sets, such as databases. It is faster than asymmetric, but the symmetric key must be kept secret and shared securely. Asymmetric uses two keys, one public (to encrypt), and one private (to decrypt it). It is more secure for data exchange over the internet such as for key exchange, or digital signatures.

1. Symmetric encryption is often used for bulk data because it is faster, and more efficient.

2. Asymmetric encryption is better for sending data securely over the internet because it uses a public key to encrypt, and a private key to decrypt. This makes it ideal for secure key exchange and digital signatures.

3. HTTPS/TLS uses both asymmetric and symmetric encryption. Asymmetric securely establishes or exchanges session key. Symmetric uses that session key to encrypt actual data.